

Isaac Han

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Research Interest

Maintaining Plasticity for Efficient Reinforcement Learning

- Maintaining plasticity to improve sample & compute efficiency of RL.
- Related: FIRE (ICLR Oral), AID (ICML)

Deep Learning Applications

- Previously explored 3D pose estimation from wearable sensors, StarCraft AI, VR haptic devices, and off-road autonomous driving.

Education

Gwangju Institute of Science and Technology (GIST)

M.S./Ph.D. Student in AI (GPA: 4.33/4.5, Advisor: Kyung-Joong Kim)

Gwangju, Korea

2020 – 2026

Bachelor's Degree Examination for Self-Education

B.S. in Computer Science (GPA: 3.7/4.3), nationally accredited by NILE

Korea

2019

Research Experience

University of Alberta

Visiting Researcher (Advisor: Rupam Mahmood)

Edmonton, Canada

Mar. 2026 – Jun. 2026

- Improving deep-RL sample efficiency by scaling the replay ratio with recent advances in streaming RL.

Publications & Preprints

FIRE: Frobenius-Isometry Reinitialization for Balancing the Stability-Plasticity Tradeoff

ICLR'26

Oral

Isaac Han, Sangyeon Park, Seungwon Oh, Donghu Kim, Hojoon Lee, Kyung-Joong Kim

- [arXiv](#) / [project page](#) / [github](#)

Activation by Interval-wise Dropout: A Simple Way to Prevent Neural Networks from Plasticity Loss

ICML'25

Sangyeon Park, Isaac Han, Seungwon Oh, Kyung-Joong Kim

- [arXiv](#)

Deep Ensemble Learning of Tactics to Control the Main Force in a Real-Time Strategy Game

Multimedia Tools and Applications (2024)

Isaac Han, Kyung-Joong Kim

- [paper](#)

Monte Carlo and Temporal Difference Methods in Reinforcement Learning [AI-eXplained]

IEEE Computational Intelligence Magazine (2023)

Isaac Han, Seungwon Oh, Hoyoun Jung, Insik Chung, Kyung-Joong Kim

- [paper](#)

Seamless-Walk: Natural and Comfortable Virtual Reality Locomotion Method with a High-Resolution Tactile Sensor

Virtual Reality (2023)

Yunho Choi, Dong-Hyeok Park, Sungha Lee, Isaac Han, Ecehan Akan, Hyeon-Chang Jeon, Yiyue Luo, Seungjun Kim, Wojciech Matusik, Daniela Rus, Kyung-Joong Kim

- [paper](#)

A New Open-Source Off-Road Environment for Benchmark Generalization of Autonomous Driving

IEEE Access (2021)

Isaac Han, Dong-Hyeok Park, Kyung-Joong Kim

- [paper](#) / [github](#) (60+ ★)

Shared Representation for 3D Pose Estimation, Action Classification, and Progress Prediction from Tactile Signals

arXiv

Isaac Han, Seoyoung Lee, Sangyeon Park, Ecehan Akan, Yiyue Luo, Joseph DelPreto, Kyung-Joong Kim

- [paper](#) / [video](#)

Recovering Plasticity of Neural Networks via Soft Weight Rescaling

arXiv

Seungwon Oh, Sangyeon Park, Isaac Han, Kyung-Joong Kim

- [arXiv](#)

Smart Insole: Predicting 3D Human Pose from Foot Pressure

2nd NeurIPS

Isaac Han, Seoyoung Lee, Sangyeon Park, Ecehan Akan, Yiyue Luo, Kyung-Joong Kim

Workshop on Touch Processing

- [paper](#)

Academic Services

Reviewer	Conference on Neural Information Processing Systems (NeurIPS)	2026
Reviewer	The ACM Conference on Human Factors in Computing Systems (CHI)	2026
Reviewer	IEEE International Conference on Robotics & Automation (ICRA)	2026
Reviewer	IEEE Conference on Games (CoG)	2025
Reviewer	IEEE Access	2023

Technical Skills

Proficient: Python, PyTorch, TensorFlow, JAX

Experienced: Docker, GitHub